

ECON 21030: Honors Econometrics

Instrumental Variables

Spring Quarter 2024

Introducing IV

So far, we’ve studied methods for estimating causal relationships that hinge on a conditional independence assumption. In the cross-sectional case, we required potential outcomes and/or structural residuals to be independent of treatment (possibly conditional on covariates). In the panel case, we’ve discussed settings such as diff-in-diff where treatment is assumed to be (conditionally) independent of changes in outcomes. But what’s a researcher to do if making a conditional independence assumption is simply a bridge too far? No matter how many excellent observables we collect, we may always be worried that unobserved omitted factors that correlate with treatment confound our estimates of causal effects. And even if you’re not, your critics almost certainly will be.

Instrumental variables (IV) provide an alternative method for estimating parameters of some models that could not be identified by conventional approaches, including OLS, propensity score methods, or other simple aggregates of differences in treated versus untreated means. These approaches work by introducing a third random variable, typically denoted Z_i , with special properties. This random variable, called an **instrument**, generally must do two things: be uncorrelated with potential confounds and be correlated with the treatment (typically called the **endogenous** variable). Besides fixed effect and diff-in-diff approaches using panel data, IV strategies are probably the single most common approach deployed in modern econometrics. There is enormous amount of richness and complexity in their application; we will only scratch the surface here.

To give a more concrete introduction, consider the following simple structure model of the returns to schooling for earnings:

$$\begin{aligned} Y_i &= \alpha + \beta_1 S_i + \beta_2 A_i + U_i \\ \text{Cov}(S_i, A_i) &\neq 0 \\ \text{Cov}(S_i, U_i) &= \text{Cov}(A_i, U_i) = 0 \end{aligned}$$

where A_i is student i ’s overall ability. If we observed both S_i and A_i , it would be straightforward to estimate β_1 using regression. Since A_i represents individual “ability” (whatever that means), however, we’re unlikely to have a good measure of it, making this strategy infeasible. As you know from our previous lectures, simply running the “short” regression of earnings on schooling generally won’t cut it:

$$Y_i = \alpha_s + \beta_s S_i + \eta_i$$

You know from omitted variables bias logic that:

$$\beta_s = \beta_1 + \beta_2 \frac{Cov(S_i, A_i)}{Var(S_i)}$$

To make progress, now suppose we also observe an indicator Z_i for whether student i was randomly selected to win a large scholarship. Consider the simplified structural model given by:

$$Y_i = \alpha + \beta_1 S_i + e_i$$

where $e_i = \beta_2 A_i + U_i$

We assume that Z_i satisfies two critical assumptions:

1. **Relevance:** $Cov(S_i, Z_i) \neq 0$
2. **Exclusion:** $Cov(e_i, Z_i) = 0$

Relevance requires that the instrument affect the endogenous variable (i.e., treatment) of interest—in this case, years of schooling. Since Z_i represents winning a massive scholarship in our running example, this seems intuitive. But one need not rest solely on assumptions here, since testing whether $Cov(S_i, Z_i) = 0$ is straightforward (**exercise: how?**).

The second assumption, exclusion, is more subtle. It requires that the instrument be uncorrelated with the structural error. In our example, this requires the instrument to be uncorrelated with ability (A_i) as well as any other unobserved determinant of earnings. It also requires that the complete structural model not include any *direct* causal effects of Z_i itself on Y_i , otherwise the term we have dubbed e_i would also include some part capturing impacts of Z_i on Y_i . Since our example scholarship was randomly assigned, we might feel fairly confident exclusion would hold here. But in general (with some exceptions) exclusion is hard to test, so you will need to make the case based on the specifics of the setting under study.

Under these two assumptions, you can easily show that:

$$Cov(Y_i, Z_i) = Cov(\alpha + \beta_1 S_i + e_i, Z_i) = \beta_1 Cov(S_i, Z_i)$$

Thus β_1 is identified by $Cov(Y_i, Z_i)/Cov(S_i, Z_i)$. Magic! But not really. The intuition here should be clear. Because Z_i has no direct or confounded relationship with Y_i itself, the only reason why Z_i should co-vary with Y_i is if Z_i shifts S_i and S_i affects Y_i . So rescaling the covariance of Y_i and Z_i by how much Z_i affects S_i should tell you something about the causal effects of S_i on Y_i .

This ratio of covariances is the **population instrumental variables coefficient**. In other words, it is an estimand. Like all estimands, it is defined by the joint distribution of the random variables it comprises and thus is defined regardless of what kind of structural model we have in mind. When we assume relevance and exclusion, however, the population IV coefficient identifies β_1 in our structural model.

Notice that we can rewrite:

$$\beta_1 = \frac{Cov(Y_i, Z_i)}{Cov(S_i, Z_i)} = \frac{Cov(Y_i, Z_i)/Var(Z_i)}{Cov(S_i, Z_i)/Var(Z_i)}$$

The numerator and denominator should look very familiar to you here. They are simply the slope coefficients on Z_i in regressions with Y_i and S_i as the outcomes, respectively. The IV estimand can therefore be expressed as the ratio of coefficients from these two regressions:

$$\begin{aligned} S_i &= \alpha_1 + \pi Z_i + \nu_i \\ Y_i &= \alpha_0 + \delta Z_i + U_i \end{aligned}$$

with $\beta_1 = \delta/\pi$. This first regression is known as the [first stage](#), with π referred to as the first stage coefficient. The second regression is known as the [reduced form](#), with δ referred to as the reduced form coefficient. Again, the IV estimand $\beta_{IV} = \delta/\pi$ is simply the ratio of the coefficients from these two regressions and always exists whenever this ratio is well-defined (i.e. $\pi \neq 0$). Only under the right assumptions about Z_i will β_{IV} identify an interesting causal parameter.

When Z_i is binary, π and δ become simple differences in means, so β_{IV} takes on a special form that is the population analogue of the [Wald estimator](#):

$$\beta_{IV} = \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[S_i|Z_i = 1] - E[S_i|Z_i = 0]}$$

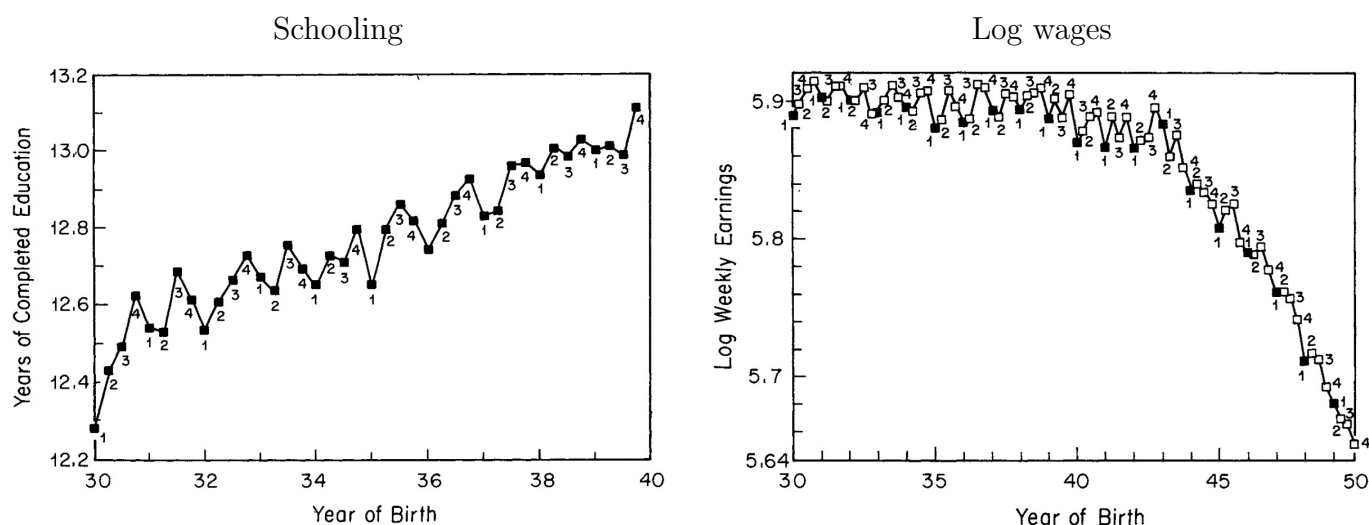
This ratio of differences-in-means is commonly used in practice, since treatments and instruments under study are often binary.

An example (Angrist and Krueger, 1991)

A simple example comes from Angrist and Krueger (1991), who were also interested in the causal effects of schooling on earnings. They noticed two important things: students can drop out of schooling when they turn 16, and students typically start school in the fall of the year where they turn 6. As a result, students born very early in the year tend to be older when they start school. They therefore reach age 16 having completed less school overall and, if they drop out, will leave schooling with less education than students born very late in the year, who cannot drop out until later.

It's clear how these facts might generate a first stage—i.e., differences in educational attainment by how early versus late in the year a student is born. Indeed, Angrist and Krueger show this is the case using Census data and treating the calendar quarter of birth as their instrument. That takes care of relevance. But what about exclusion? What would you need to believe about the timing of quarter of birth for it to serve as a valid instrument in this case? I leave this question as an [exercise](#).

Regardless, here are the two key figures from the paper:



It seems clear that birth timing affects schooling and log wages. Constructing Wald estimates using a binary instrument Z_i defined as an indicator for being born in the first quarter of the year, they find sizable returns to education. Indeed, the returns are larger than what one would get with a simple OLS regression of schooling on education alone.

Panel B: Wald Estimates for 1980 Census—Men Born 1930–1939

	(1) Born in 1st quarter of year	(2) Born in 2nd, 3rd, or 4th quarter of year	(3) Difference (std. error) (1) – (2)
ln (wkly. wage)	5.8916	5.9027	–0.01110 (0.00274)
Education	12.6881	12.7969	–0.1088 (0.0132)
Wald est. of return to education			0.1020 (0.0239)
OLS return to education			0.0709 (0.0003)

a. The sample size is 247,199 in Panel A, and 327,509 in Panel B. Each sample consists of males born in the United States who had positive earnings in the year preceding the survey. The 1980 Census sample is drawn from the 5 percent sample, and the 1970 Census sample is from the State, County, and Neighborhoods 1 percent samples.

b. The OLS return to education was estimated from a bivariate regression of log weekly earnings on years of education.

Incorporating covariates

As with our work on unconfoundedness, we may be willing to make relevance and (more often) exclusion assumptions only conditional on some covariates X_i . In the Angrist and Krueger case, for example, quarter of birth is correlated with age, so we may only want to make log earnings

comparisons conditional on age. We can do so by augmenting our model to incorporate these controls:

$$Y_i = \alpha + \beta_1 S_i + X_i' \gamma + v_i$$

The extra covariates X_i are referred to as **exogenous covariates** here to distinguish them from the **endogenous regressor** S_i , the coefficient on which we're trying to use our instrument to identify. We typically don't want to interpret the parameter γ as causal. The purpose of these controls is just to soak up any reason why Z_i may not satisfy exclusion. Thus we typically also assume that γ is defined such that $Cov(X_i, v_i) = 0$.

We can now proceed exactly as we did before with a first stage and reduced form given by:

$$\begin{aligned} S_i &= \alpha_1 + \pi Z_i + X_i' \gamma_1 + \nu_i \\ Y_i &= \alpha_0 + \delta Z_i + X_i' \gamma_0 + U_i \end{aligned}$$

The IV estimand is still defined as δ/π , which we know by FWL can be written as:

$$\beta_{IV} = \frac{Cov(Y_i, \tilde{Z}_i)}{Cov(S_i, \tilde{Z}_i)}$$

where $\tilde{Z}_i$ is the residual from a regression of Z_i on X_i .

Whether β_{IV} identifies the parameter of interest β_1 can be interpreted in two ways. We can either go back to the assumptions made earlier in the binary case but treating $\tilde{Z}_i$ as the instrument, or we can assume that $\pi \neq 0$ and $Cov(Z_i, v_i) = 0$. Either way, we are assuming that relevance and exclusion holds only *conditional* on these controls.

Multiple instruments

Suppose now that we want to use *multiple* instruments. In the Angrist and Krueger case, for example, we may want to use an indicator for being born in each quarter of the year as opposed to just using the contrast between people born in the first quarter versus later. Since we could construct an estimate of β_1 using these instruments in multiple ways (e.g., comparing first versus second, second versus third, etc.), we say that β_1 is **over-identified**.

What should we do? Consider a regression of S_i onto all the available instruments:

$$S_i = \alpha_1 + Z_i' \pi + X_i' \gamma_1 + \nu_i$$

Now plug this definition of S_i back into our structural model

$$\begin{aligned} Y_i &= \alpha + \beta_1(\alpha_1 + Z_i' \pi + X_i' \gamma_1 + \nu_i) + X_i' \gamma + v_i \\ &= \alpha + \beta_1(\alpha_1 + Z_i' \pi + X_i' \gamma_1) + X_i' \gamma + \beta_1 \nu_i + v_i \\ &= \alpha + \beta_1 \hat{S}_i + X_i' \gamma + \underbrace{\beta_1 \nu_i + v_i}_{\xi_i} \end{aligned}$$

Notice that $\hat{S}_i$ is uncorrelated with ξ_i by assumption and construction. Assumption implies that it is uncorrelated with v_i , since it is a linear function of Z_i and X_i and they are both uncorrelated with v_i . Construction implies it is uncorrelated with ν_i , since that is the residual from a regression of S_i onto Z_i and X_i .

This suggests that simple OLS estimation of a regression of Y_i onto $\hat{S}_i$ and X_i would recover β_1 . This estimator is known as the [the two-stage least squares](#) (2SLS) estimator. In some cases, it is also the most efficient way to use the information contained in all of your instruments.

2SLS can be used in practice by following this two-step procedure. One first regresses the endogenous variable (e.g., schooling) onto the instruments and controls. Then one computes the fitted values. Finally, one regresses the outcome onto these fitted values and the exogenous controls, a regression known as the [second stage](#). Remembering to include the controls in last part is important! If the exogenous controls X_i are used in the first stage, then they also must be used in the second stage. Otherwise, you are in fact treating the X_i as additional instruments ([exercise: why?](#)).

2SLS only differs from the approaches we studied above when $\dim(Z_i) > 1$. I leave it as a [final exercise](#) to show that in the simple case where Z_i is scalar random variable, 2SLS and estimands such as the Wald estimand are equivalent.

Estimation

So far, we've been studying properties of IV in the population. Suppose now we have an actual sample of N observations of (Y_i, S_i, Z_i, X_i) and want to try to estimate some parameters using the ideas we've just discussed. One option would be to construct sample analogues of the covariances that define the IV estimands we discussed above. But most practitioners and software focus on 2SLS, so we'll focus on that here too.

Suppose $\dim(X_i) = L$ (including a constant), $\dim(Z_i) = J$, and $\dim(S_i) = K$, with $J \geq K$ (more on this in a second). Let $\mathbf{X}$ be the N by $L + K$ matrix with (X'_i, S'_i) on the i th row. Let $\mathbf{Z}$ be the N by $L + J$ matrix with (X'_i, Z'_i) on the i th row. As usual, we let $\mathbf{Y}$ be the N by one vector of outcomes, and $\mathbf{U}$ be the N by one vector of structural residuals, so that we can write our structural model as:

$$\mathbf{Y} = \mathbf{X}\beta + \mathbf{U}$$

where $\beta = (\alpha, \gamma', \beta_1)'$ contains the coefficients of the structural model.

Using 2SLS requires regressing the outcome onto the first stage fits. These fits are given by:

$$\hat{\mathbf{X}} = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X} = \mathbf{P}_\mathbf{Z}\mathbf{X}$$

Notice that $\hat{\mathbf{X}}$ includes the fitted values both for the endogenous variables S_i and the exogenous controls X_i . Since X_i is also included in $\mathbf{Z}$, however, we know that predicted values for the latter satisfy $\hat{X}_i = X_i$ ([exercise: prove this](#)). This is therefore equivalent to including X_i in the second stage.

Our 2SLS estimator can thus be written as the OLS regression of Y_i on $\hat{X}_i$:

$$\begin{aligned}\hat{\beta}_{2SLS} &= (\hat{\mathbf{X}}' \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}' \mathbf{Y} \\ &= (\mathbf{X}' \mathbf{P}_Z \mathbf{X})^{-1} \mathbf{X}' \mathbf{P}_Z \mathbf{Y} \\ &= (\hat{\mathbf{X}}' \mathbf{X})^{-1} \hat{\mathbf{X}}' \mathbf{Y}\end{aligned}$$

This is the general form of 2SLS. In the event that $J = K$, so that we have exactly as many instruments as endogenous variables, because $\mathbf{Z}$ and $\mathbf{X}$ have the same dimension one can also show that:

$$\begin{aligned}(\hat{\mathbf{X}}' \mathbf{X})^{-1} &= (\mathbf{X}' \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1} \mathbf{Z}' \mathbf{X})^{-1} \\ &= (\mathbf{Z}' \mathbf{X})^{-1} (\mathbf{Z}' \mathbf{Z}) (\mathbf{X}' \mathbf{Z})^{-1}\end{aligned}$$

which implies that:

$$\begin{aligned}\hat{\beta}_{2SLS} &= (\mathbf{Z}' \mathbf{X})^{-1} (\mathbf{Z}' \mathbf{Z}) (\mathbf{X}' \mathbf{Z})^{-1} \mathbf{X}' \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1} \mathbf{Z}' \mathbf{Y} \\ &= (\mathbf{Z}' \mathbf{X})^{-1} \mathbf{Z}' \mathbf{Y}\end{aligned}$$

This is the “traditional” IV estimator, because it can also motivated from the exclusion assumption that $E[\mathbf{Z}' \mathbf{U}] = 0 = E[\mathbf{Z}' (\mathbf{Y} - \mathbf{X} \beta)]$, so that $\beta = E[\mathbf{Z}' \mathbf{X}]^{-1} E[\mathbf{Z}' \mathbf{Y}]$ whenever this inverse exists (which requires $J = K$). You can see that this simple form of estimator is just a ratio of sample covariances, as discussed above.

When $J \geq K$, so that β is **over-identified**, 2SLS cannot be written in this simple form. But in the **just-identified case** (meaning $J = K$), 2SLS and IV coincide. In fact, when $J = K$ so would any estimator of the form $(\mathbf{Z}' \mathbf{A} \mathbf{X})^{-1} \mathbf{Z}' \mathbf{A} \mathbf{Y}$, where $\mathbf{A}$ is an invertible $L+J$ by $L+K$ matrix. Intuitively, in the just-identified case there is only one way to solve the population moment restrictions that identify β , so many estimators are going to be identical.

When $J < K$, we say that the system is **under-identified**. 2SLS will not work in this case. To see why, recall that estimating $\hat{\beta}_{2SLS}$ requires $\hat{\mathbf{X}}' \hat{\mathbf{X}}$ to be invertible. That requires $\hat{\mathbf{X}}$ to have full column rank. Because each column of $\hat{\mathbf{X}}$ is a linear combination of the $\mathbf{Z}$, this cannot happen when $J < K$. For example, in the simple case where $K = 1$ and $J = 0$, so that we have no instruments at all, then $\hat{S}_i$ is a linear combination of X_i alone, but the X_i are included in the second-stage regression as well, so there is a clear colinearity problem.

Inference

What about standard errors? Using the same ideas we used to study the asymptotic distribution of OLS, notice that we can write:

$$\begin{aligned}\hat{\beta}_{2SLS} &= (\hat{\mathbf{X}}' \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}' \mathbf{Y} \\ &= (\hat{\mathbf{X}}' \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}' (\mathbf{X} \beta + \mathbf{U}) \\ &= \beta + (\hat{\mathbf{X}}' \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}' \mathbf{U} \\ \rightarrow \hat{\beta}_{2SLS} - \beta &= (\hat{\mathbf{X}}' \hat{\mathbf{X}})^{-1} \mathbf{X}' \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1} \mathbf{Z}' \mathbf{U} \\ &= \left(\frac{\hat{\mathbf{X}}' \hat{\mathbf{X}}}{N} \right)^{-1} \left(\frac{\mathbf{X}' \mathbf{Z}}{N} \right) \left(\frac{\mathbf{Z}' \mathbf{Z}}{N} \right)^{-1} \left(\frac{\mathbf{Z}' \mathbf{U}}{N} \right)\end{aligned}$$

The first three terms will converge to their probability limits by the law of large numbers under mild conditions. We will use the notation $V_{AB} = E[A'B/N]$ to denote these limits. The final term, $\mathbf{Z}'\mathbf{U}/N = \frac{1}{N} \sum_{i=1}^N Z_i U_i$. The central limit theorem tells us that, rescaled by $\sqrt{N}$, this object converges to a normal with variance given by the variance of $Z_i U_i$.

Putting everything together, we therefore have:

$$\begin{aligned} \sqrt{N} \left(\hat{\beta}_{2SLS} - \beta \right) &\xrightarrow{d} N(0, \Sigma) \\ \Sigma &= V_{\hat{\mathbf{X}}\hat{\mathbf{X}}}^{-1} V_{\mathbf{X}\mathbf{Z}} V_{\mathbf{Z}\mathbf{Z}}^{-1} \text{Var}(Z_i U_i) V_{\mathbf{Z}\mathbf{Z}}^{-1} V_{\mathbf{X}\mathbf{Z}} V_{\hat{\mathbf{X}}\hat{\mathbf{X}}}^{-1} \end{aligned}$$

Just like in the OLS case, we can now proceed to construct an estimator of Σ using the plug-in versions of V_{AB} and using $\mathbf{Z}'\mathbf{D}\mathbf{Z}/N$, where $\mathbf{D} = \text{diag}(\hat{\mathbf{U}})$, as an estimator of $\text{Var}(Z_i U_i)$. Thus:

$$\begin{aligned} \hat{\Sigma} &= \left(\frac{\hat{\mathbf{X}}'\hat{\mathbf{X}}}{N} \right)^{-1} \left(\frac{\mathbf{X}'\mathbf{Z}}{N} \right) \left(\frac{\mathbf{Z}'\mathbf{Z}}{N} \right)^{-1} \left(\frac{\mathbf{Z}'\mathbf{D}\mathbf{Z}}{N} \right) \left(\frac{\mathbf{Z}'\mathbf{Z}}{N} \right)^{-1} \left(\frac{\mathbf{X}'\mathbf{Z}}{N} \right) \left(\frac{\hat{\mathbf{X}}'\hat{\mathbf{X}}}{N} \right)^{-1} \\ &= N(\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1} \mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1} \mathbf{Z}'\mathbf{D}\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1} \mathbf{X}'\mathbf{Z}(\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1} \end{aligned}$$

As usual, if we want to use $\hat{\Sigma}$ to construct an estimate of the sampling variance of $\hat{\beta}$, we have to undo the normalization by $\sqrt{N}$, which will kill the leading N in the second line of the previous display.

Like with OLS, the asymptotic variance of 2SLS simplifies significantly when errors are homoskedastic, implying that $\text{Var}(Z_i U_i) = \sigma^2 E[Z_i Z_i']$. This implies that:

$$\begin{aligned} \Sigma_{hom} &= \sigma^2 V_{\hat{\mathbf{X}}\hat{\mathbf{X}}}^{-1} V_{\mathbf{X}\mathbf{Z}} V_{\mathbf{Z}\mathbf{Z}}^{-1} V_{\mathbf{X}\mathbf{Z}} V_{\hat{\mathbf{X}}\hat{\mathbf{X}}}^{-1} \\ &= \sigma^2 V_{\hat{\mathbf{X}}\hat{\mathbf{X}}}^{-1} \end{aligned}$$

where the second line follows from the fact that $(\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1} = (\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X})^{-1}$. This expression has the simple interpretation as the residual structural variance divided by the variance of the first-stage fits. All else equal, more variance in fits generates a tighter standard error. Like OLS in the homoskedastic case, 2SLS here is also the efficient linear IV estimator. In the over-identified case, however, other estimators may in fact do better under heteroskedasticity.

Although the 2SLS estimate of β can be obtained from manually regressing Y_i on $\hat{S}_i$ and X_i , it is generally unwise to do so because your standard errors will be incorrect. To see why, consider the simple bivariate case with homoskedastic standard errors. If you use “manual” 2SLS and estimate:

$$Y_i = \alpha + \beta \hat{S}_i + V_i$$

The variance of $\hat{\beta}$ will be $\text{Var}(V_i)/\text{Var}(\hat{S}_i)$. This is the variance of the second stage residuals. The structural model, however, says that $Y_i = \alpha + \beta S_i + U_i$, so $V_i = U_i + \beta(S_i - \hat{S}_i)$. The variance of V_i will thus typically be too big. Instead, you want to use the structural residuals, which can be estimated as $Y_i - \hat{\alpha} - \hat{\beta} S_i$ (not $-\hat{\beta} \hat{S}_i$).

Heterogeneous effects

Throughout the previous few sections, we were studying structural models that implied constant causal effects. To conclude, we'll study the properties of IV in a world with potentially heterogeneous effects. This framework uses potential outcomes and is the predominate framework used to think about IV in modern applied work. As we will see, however, though the interpretation of some estimands may change, the tools used to estimate them and conduct inference will not.

To begin, let our outcome continue to be Y_i , define a binary treatment T_i , and let a binary instrument be denoted Z_i . Potential outcomes, $Y_i(t, z)$, are indexed by both T_i and Z_i , allowing the value of the treatment *and* the value of the instrument to influence outcomes. We also consider [potential treatments](#), $T_i(z)$, which also depend on the instrument. In our college scholarship example from before, you can imagine Y_i is earnings, T_i is attending college, and Z_i is the scholarship. Our notation so far allows both attending college and getting the scholarship to affect outcomes directly, and allows getting the scholarship to affect the decision to attend college. As usual, we observe only one potential outcome and treatment given by $T_i(Z_i)$ and $Y_i(T_i, Z_i)$, respectively.

The first assumption we need to make progress is [independence](#):

$$\text{Independence : } (Y_i(0, 0), Y_i(1, 0), Y_i(0, 1), Y_i(1, 1), T_i(0), T_i(1)) \perp\!\!\!\perp Z_i$$

This assumption says that all potential outcomes are independent of Z_i . It can be guaranteed by using a randomization protocol for Z_i , or argued that it holds because of facts about the world / setting. We also need an [exclusion](#) restriction:

$$\text{Exclusion : } Y_i(t, 1) = Y_i(t, 0) \forall t, i$$

Exclusion says that only treatment can effect outcomes, not Z_i . In our scholarship example, it would require that the scholarship only change log-earnings *through* the decision to go to school. It cannot also change log earnings directly (e.g., by reducing the amount of debt you take on, and therefore how much you work).

Together these assumptions deliver what we called exclusion in the constant effect case. Independence is equivalent to assuming that $Cov(e_i, Z_i) = 0$. And exclusion is equivalent to assuming that Z_i does not directly enter the structural model. In the heterogeneous effect world, exclusion and independence are often simply combined into:

$$\text{Exclusion/independence : } (Y_i(0), Y_i(1), T_i(0), T_i(1)) \perp\!\!\!\perp Z_i$$

We also need a version of the [relevance](#) assumption in this setting. That can be accomplished by assuming that $Pr(T_i = 1|Z_i = 1) > Pr(T_i = 1|Z_i = 0)$ (or vice-versa). As in the constant effect case, this assumption is easily testable.

Finally, we typically also need to assume [monotonicity](#), which requires that $T_i(1) \geq T_i(0) \forall i$ (or vice-versa). This assumption implies that instruments only induce “one-way flows,” making every individual weakly more likely to take-up a treatment. Monotonicity is difficult to test, since as per usual we only ever get to see one of $T_i(1)$ and $T_i(0)$ for each person, but not both.

Under monotnicity, the world can be broken down into four groups of people:

1. [Never takers](#) with $T_i(0) = T_i(1) = 0$, who would not be treated regardless of the value of the instrument.

2. **Always takers** with $T_i(0) = T_i(1) = 1$, who would be treated regardless of the value of the instrument.
3. **Compliers** with $T_i(1) > T_i(0)$, whose treatment status depends on the instrument.

People with $T_i(1) < T_i(0)$ are known as **defiers** and are ruled out by monotonicity. This terminology comes to us from randomized trials, where subjects assigned to treatment ($Z_i = 1$) may or may not comply (i.e., $T_i = 1$). **Perfect compliance** refers the situation where every subject has to comply and cannot access treatment without being assigned to it, so that $Pr(T_i(1) = T_i(0) = 1) = Pr(T_i(1) = T_i(0) = 0) = 0$.

Now let's revisit the Wald estimand we studied before using these assumptions. The numerator can be written as:

$$\begin{aligned}
E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] &= E[Y_i(T_i(1))|Z_i = 1] - E[Y_i(T_i(0))|Z_i = 0] \\
&= E[Y_i(T_i(1)) - Y_i(T_i(0))] \quad \text{by independence} \\
&= E[Y_i(1) - Y_i(0)|T_i(1) > T_i(0)]Pr(T_i(1) > T_i(0)) \\
&\quad - E[Y_i(1) - Y_i(0)|T_i(1) < T_i(0)]Pr(T_i(1) < T_i(0)) \quad \text{by exclusion} \\
&= E[Y_i(1) - Y_i(0)|T_i(1) > T_i(0)]Pr(T_i(1) > T_i(0)) \quad \text{by monotonicity}
\end{aligned}$$

The denominator can likewise be written as:

$$\begin{aligned}
E[T_i|Z_i = 1] - E[T_i|Z_i = 0] &= E[T_i(1)|Z_i = 1] - E[T_i(0)|Z_i = 0] \\
&= E[T_i(1) - T_i(0)] \quad \text{by independence} \\
&= Pr(T_i(1) > T_i(0)) - Pr(T_i(1) < T_i(0)) \\
&= Pr(T_i(1) > T_i(0)) \quad \text{by monotonicity}
\end{aligned}$$

Thus our simple Wald estimand delivers:

$$\frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[T_i|Z_i = 1] - E[T_i|Z_i = 0]} = E[Y_i(1) - Y_i(0)|T_i(1) > T_i(0)]$$

This object is called a **local average treatment effect (LATE)**. It captures the effects of treatment for the population *induced* into treatment by the instruments. Always and never takers don't contribute because their treatment status is not affected by Z_i , and hence their Y_i s are also unaffected. Only compliers cause the distribution of Y_i to differ between $Z_i = 1$ and $Z_i = 0$, and it's their average treatment effect that we capture with IV.

The LATE is not the ATE. Compliers may be special and weird in some ways that make the treatment effects very different from those of the full population. Consider the Angrist and Krueger example from earlier, for example. Who are the compliers, and how might they differ from the overall population of students? Nevertheless, the LATE theorem shows you that this is what is estimated by IV models under heterogeneous effects. You may wish you had a more externally valid estimate, but this is as good as you can do with simple estimators like we've discussed here.

A natural extension is to ask what happens if we wish to use the model but only assume that potential outcomes and treatments are conditionally independent and excludable, so that:

$$(Y_i(0), Y_i(1), T_i(0), T_i(1)) \perp\!\!\!\perp Z_i | X_i$$

As was the case with conditional unconfoundedness, we could estimate X_i -specific LATES as:

$$\frac{E[Y_i|Z_i = 1, X_i = x] - E[Y_i|Z_i = 0, X_i = x]}{E[S_i|Z_i = 1, X_i = x] - E[S_i|Z_i = 0, X_i = x]} = E[Y_i(1) - Y_i(0)|T_i(1) > T_i(0), X_i = x] = LATE(x)$$

We could then average these up however we liked. If we estimate 2SLS using a saturated set of indicators for values of X_i (including a constant), which I'll denote W_i , interacted with Z_i , so that the first and second stages are respectively given by:

$$\begin{aligned} T_i &= W_i' \alpha_0 + Z_i W_i' \beta_0 + \eta_i \\ Y_i &= W_i' \alpha_1 + \beta_1 \hat{T}_i + \epsilon_i \end{aligned}$$

Then the 2SLS estimate of β_1 recovers a weighted average of $LATE(x)$ s:

$$\begin{aligned} \beta_1 &= E[\omega(X_i) LATE(X_i)] \\ \omega(X_i) &= \frac{Var(E[T_i|X_i, Z_i]|X_i)}{E[Var(E[T_i|X_i, Z_i]|X_i)]} \end{aligned}$$

Like OLS, 2SLS is putting more weight on cells where there is more variation to estimate $LATE(x)$. In OLS, we get more weight in X_i cells where there is a lot of variance in treatment. In 2SLS, we get more weight in X_i cells where there is a lot of variance in expected treatments given X_i and Z_i —that is, $\hat{T}_i$. As we showed above, the variance of 2SLS is inversely proportional to the variance of $\hat{T}_i$ under homoskedasticity, so these are the covariate-specific LATEs that are estimated most accurately.

As noted earlier, the heterogeneous effect assumptions, which are collectively often called the [LATE framework](#), change the interpretation of estimands but not the tools / mechanics. You can still run 2SLS and construct standard errors as we discussed earlier. All that changes is how you talk about what you learn from them!

There's lots more to learn here, including how to characterize compliers, heterogeneous effects interpretations of 2SLS using multiple instruments and continuous/ordered treatments and instruments, using multiple instruments to develop tests of exclusion, what can go wrong with non-saturated controls, and more. Unfortunately we have run out of runway here, so we'll have to save these topics for another quarter!